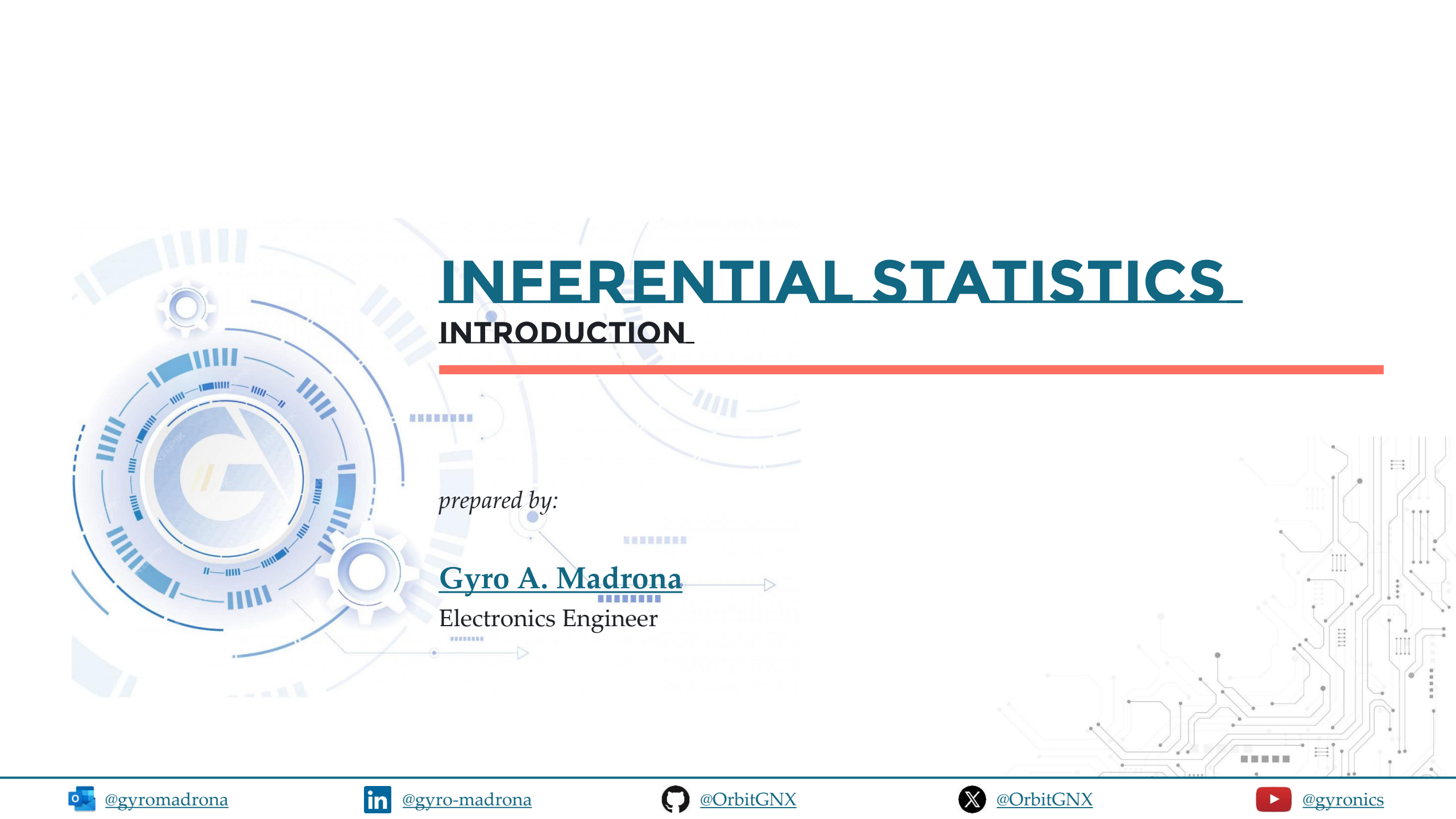




INFERENCE STATISTICS

INTRODUCTION

prepared by:

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Electronics Engineer

TOPIC OUTLINE

Inferential Statistics

Distribution

Histogram

Normal Distribution



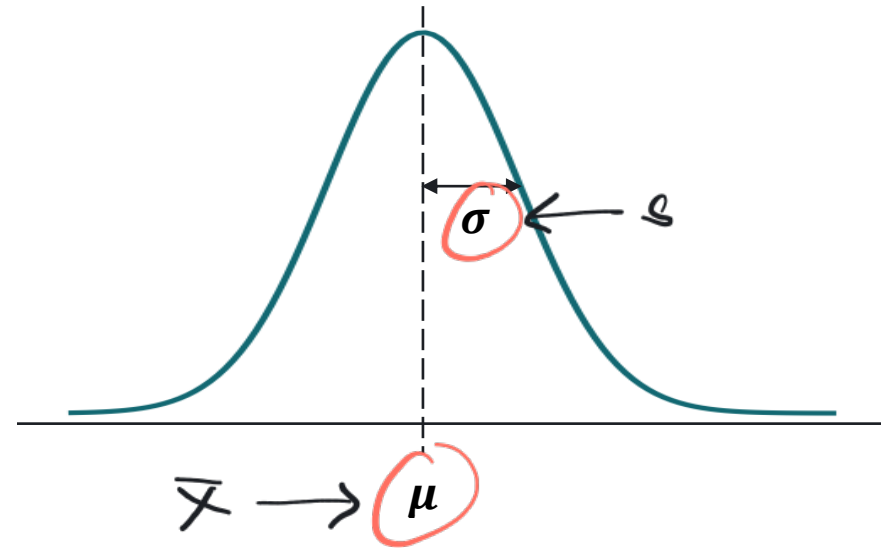
DISTRIBUTION



INFERENCE STATISTICS

Inferential statistics is a branch of statistics that uses **sample** data to draw conclusions (inferences) about a population.

Normal Distribution



sample represents population



DISTRIBUTION

Distribution or the probability distribution describes the probabilities or frequencies of different outcomes in an experiment or observed data.

Rolling one die



Outcome	Probability
1	1/6 or 0.17
2	1/6 or 0.17
3	1/6 or 0.17
4	1/6 or 0.17
5	1/6 or 0.17
6	1/6 or 0.17
7 (all else)	0

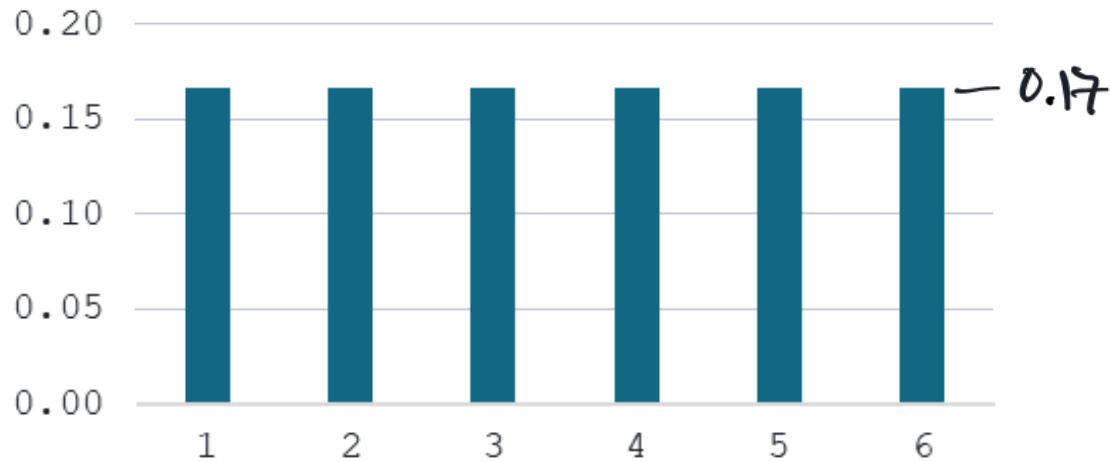
Sum of probabilities = 1 or 100%



UNIFORM DISTRIBUTION

Discrete data

One die distribution



Rolling one die



Outcome	Probability
1	1/6 or 0.17
2	1/6 or 0.17
3	1/6 or 0.17
4	1/6 or 0.17
5	1/6 or 0.17
6	1/6 or 0.17
7 (all else)	0

Sum of probabilities = 1 or 100%



DISTRIBUTION

36 possible outcomes

(1,1) (2,1) (3,1) (4,1) (5,1) (6,1)
(1,2) (2,2) (3,2) (4,2) (5,2) (6,2)
(1,3) (2,3) (3,3) (4,3) (5,3) (6,3)
(1,4) (2,4) (3,4) (4,4) (5,4) (6,4)
(1,5) (2,5) (3,5) (4,5) (5,5) (6,5)
(1,6) (2,6) (3,6) (4,6) (5,6) (6,6)

$\frac{6}{36}$

$\frac{1}{36}$

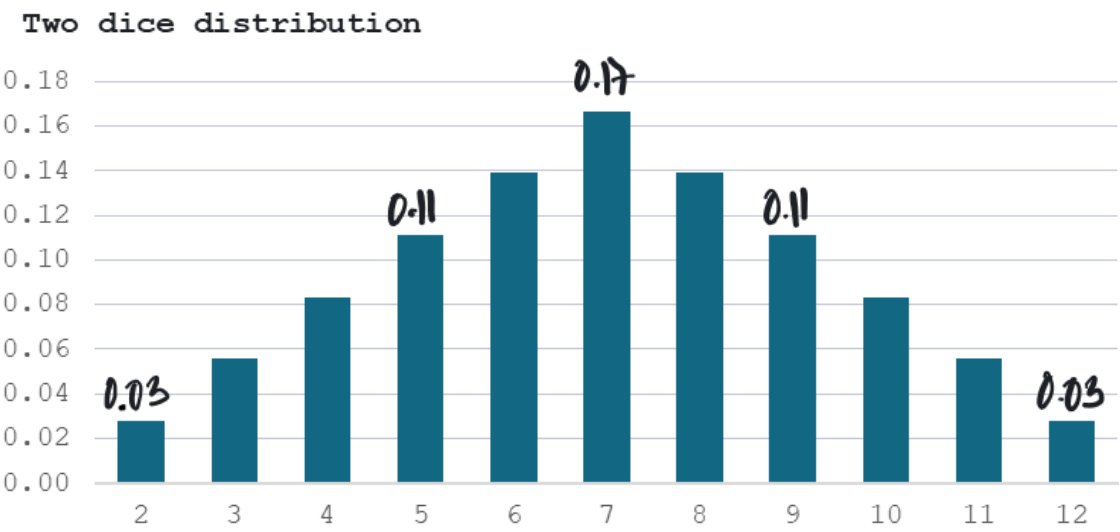
Rolling two dice

Sum	Probability
2	0.03
3	0.06
4	0.08
5	0.11
6	0.14
7	0.17
8	0.14
9	0.11
10	0.08
11	0.06
12	0.03
All else	0



DISTRIBUTION

Discrete data



Rolling two dice



Sum	Probability
2	0.03
3	0.06
4	0.08
5	0.11
6	0.14
7	0.17
8	0.14
9	0.11
10	0.08
11	0.06
12	0.03
All else	0

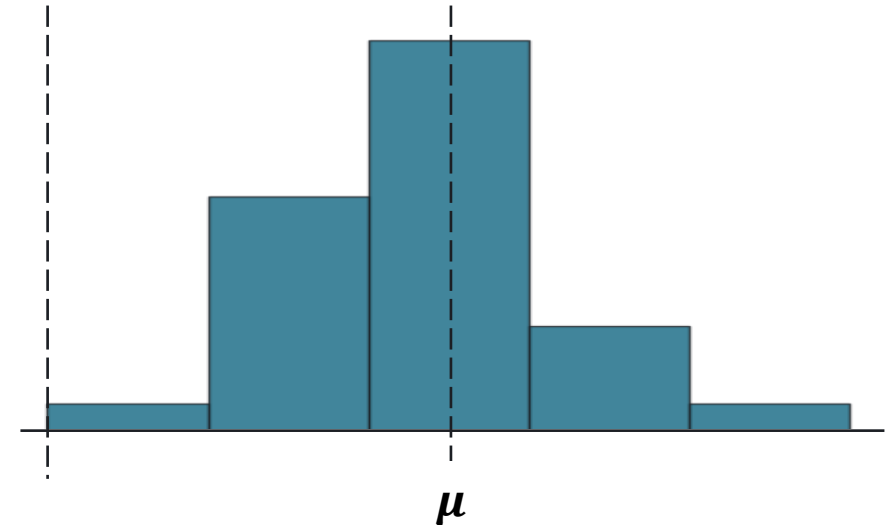


HISTOGRAM

Histograms are used to visualize the shape, spread, and central tendency of data, making them a useful tool for assessing whether a dataset follows a normal distribution or deviates from it.

Histogram

Frequency



syntax

```
import matplotlib.pyplot as plt  
  
plt.hist(data, bins = 10)  
  
plt.show()
```



NORMAL DISTRIBUTION



NORMAL DISTRIBUTION

A normal distribution is a probability distribution where the values of a random variable are distributed symmetrically. Also known as Gaussian distribution or bell curve because of its shape.



Johann Carl Friedrich Gauss



NORMAL DISTRIBUTION

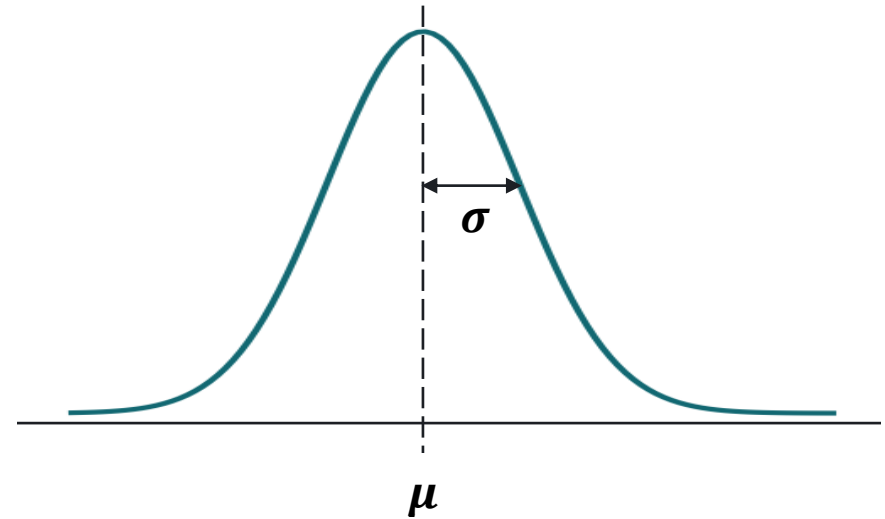
A normal distribution is a probability distribution where the values of a random variable are distributed symmetrically. Also known as Gaussian distribution or bell curve because of its shape.

Denoted by

$$N(\mu, \sigma^2)$$

↑ variance

Bell Curve



mean = median = mode



NORMAL DISTRIBUTION

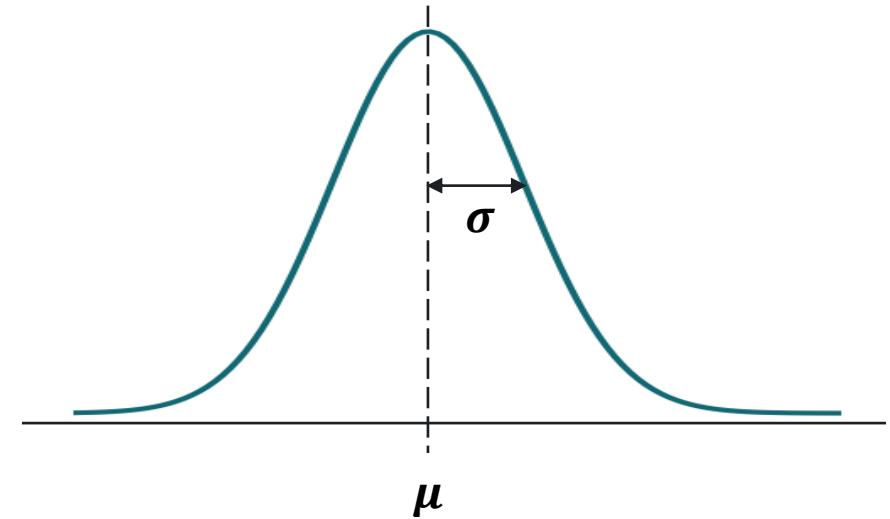
A normal distribution is a probability distribution where the values of a random variable are distributed symmetrically. Also known as Gaussian distribution or bell curve because of its shape.

Formula

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Bell Curve

reference in determining
normal vs non-normal data



mean = median = mode

syntax

```
p = stats.norm.pdf(x, mean, std)
```

```
plt.plot(x, p)
```

```
plt.show()
```

EXERCISE

The dataset consists of 30 samples of current measurements (in mA). Generate a **normal distribution plot** in a Jupyter Notebook.

dataset

current-test.csv

Current	Response
Sample	Current
1	12.0
2	15.0
3	8.3
4	9.7
5	12.0
6	13.9
7	14.1
8	9.2
9	12.4
10	13.7
11	10.6
12	21.5
13	12.0



LABORATORY

